



SILVER OAK UNIVERSITY

Silver Oak College of Computer Application

Master of Computer Application

Course Name: Core Java
Course Code: CAM5001C
Semester: 1st

Prerequisite:

1. Basic understanding of programming concepts like variables, control structures, and object oriented principles. Familiarity with a development environment and basic problem-solving skills will also be essential.

Course Objectives:

1. To understand the fundamentals of Java programming including syntax, data types, operators, and control structures.
2. To develop object-oriented programming skills using classes, objects, inheritance, polymorphism, abstraction, and encapsulation.
3. To implement exception handling, multithreading, and file handling concepts for building robust Java applications.
4. To gain knowledge of Java collections, packages, interfaces, and GUI programming for developing interactive applications.
5. To design and develop real-world applications using Core Java concepts and standard libraries.

Teaching Scheme:

Teaching & Learning Scheme (In Hrs Per Semester)									Total Credits
Classroom Instruction						Extended Learning		Total Hours	
L		T		P					
Cr	Hr	Cr	Hr	Cr	Hr	TW	SL	Hr	C
4	60	0	0	0	0	40	20	120	4

Contents:

Unit	Topics	Teaching Hours	% Weightage
1	Introduction to Java: Principles of Object-oriented language, Java Essentials, Java Virtual Machine, Program Structure ,Variables, Primitive Data Types, Identifier, Literals, Operators, Flow of Control, Classes and Objects, Constructors, Class Variable and Methods – Static Keyword, this keyword, Arrays, One-dimensional arrays, Two dimensional arrays, Command line arguments	15	25%
2	Inheritance and Interface: Inheritance, Types of Inheritance Defining subclasses ,super keyword, Method overloading ,Method overriding ,Polymorphism, Final keyword, Abstract class, interfaces ,inheritance in interface Object classes,	15	25%

	package and import statements, Access specifies ,java. Lang package, String, String Buffer Wrapper classes, Math class.		
3	Exception Handling & Multithreading: Exception Introduction, Exception Handling Techniques, User Defined Exception, The Throwable class, Exception chaining, Handling exceptions with try and catch , Use of the finally block, Multithreading in Java, java.lang.Thread, Main Thread, Creation of new Threads, Thread State in Java	15	25%
4	Generic Programming: Why Generic Programming? ,Generic Class , Generic Methods , Generic Code and the Virtual Machine, Restrictions and Limitations, Inheritance Rules for Generic Types , WildCard Types classes from java.util package, Date, TimeZone, Calendar and the Gregorian Calendar classes, Collection Framework, Collection interface , Set and List interfaces , Map Interface , Generics in the Collection Framework , Regular Expressions, Pattern and Matcher classes,Varargs and the Formatter class	15	25%
		60	100%

Course Outcomes (For Theory):

CO No.	CO Statement	Blooms Taxonomy Level	Unit
CO-1	Explain core Java concepts including object-oriented principles, program structure, data types, control flow, classes, objects, constructors, arrays, and command-line arguments.	Understand (L2)	1
CO-2	Apply inheritance, polymorphism, interfaces, packages, strings, wrapper classes, and related concepts to develop reusable object-oriented Java programs.	Apply (L3)	2
CO-3	Apply exception handling and multithreading concepts to develop robust and concurrent Java applications.	Apply (L3)	3
CO-4	Apply generics, collections, utility classes, regular expressions, and related Java features to develop flexible and reusable programs.	Understand (L3)	4

Course Outcomes (For Practical): N.A

CO No.	CO Statement	Blooms Taxonomy Level	Unit	Practical Cluster

Teaching & Learning Pedagogy approaches:

Chalk and Talk Method
PPT-based Teaching
Demonstration Method
Hands-on Practical Sessions
Interactive Classroom Discussion
Problem Solving and Coding Practice
Assignment and Mini Project Based Learning
Case Study Based Learning
E-learning Resources and Video Tutorials

List of Tutorials: N.A**List of Experiments: N.A****Major Equipment and Simulation Tools:**

1. Modern Desktop Computer System with Updated Operating System
2. Minimum 4 GB RAM
3. Java Development Kit (JDK 8 or Above)
4. Apache NetBeans / Eclipse IDE / IntelliJ IDEA
5. Java Applet Viewer / Web Browser Support for Applets
6. Projector and Smart Classroom Facilities

Books Recommended:

1. Herbert Schildt, Java: The Complete Reference, 13th Edition, McGraw-Hill, 2025.
2. Cay S. Horstmann, Core Java Volume I – Fundamentals, 13th Edition, Pearson, 2025.
3. Paul Deitel & Harvey Deitel, Java How to Program, 12th Edition, Pearson, 2023.
4. Kathy Sierra & Bert Bates, Head First Java, 3rd Edition, O'Reilly, 2022.

Term Work Module:**40 HOURS****Objective:**

To develop practical knowledge of Java programming concepts through hands-on exercises, problem-solving, mini applications, multithreading, exception handling, and applet-based programs.

Term Work	Statement	Hours	CO No.
Term Work 1	Innovative Assignment: Create a creative presentation or infographic illustrating the history of Java, its features, JVM architecture, and OOP concepts. Develop simple Java programs demonstrating variables, data types,	10 Hours	CO1

	operators, classes, objects, constructors, and command-line arguments, and submit a brief report with screenshots of outputs.		
Term Work 2	Innovative Assignment: Design a Java-based mini application that demonstrates the use of arrays, inheritance, abstract classes, and interfaces. Include programs showcasing static and this keywords, different types of arrays, method overloading/overriding, and interface implementation, along with a short report explaining the concepts and outputs.	10 Hours	CO2
Term Work 3	Innovative Assignment: Develop a Java project that demonstrates the creation and use of packages, string manipulation using String and StringBuffer classes, wrapper classes, and exception handling techniques including user-defined exceptions. Submit the source code with outputs and a brief report explaining the implementation of each concept.	10 Hours	CO3, CO4
Term Work 4	Innovative Assignment: Smart Collection Manager: Develop a simple real-life Java mini application such as a Student Record Manager, Contact Book, Library Book Manager, or Product Manager using Generics and the Collection Framework. Demonstrate a generic class or generic method and use suitable List, Set, or Map collections. Include at least one utility concept such as Date/Calendar, Regular Expressions, Pattern/Matcher, Varargs, or Formatter. Submit source code, sample outputs, screenshots, and a brief report explaining the concepts used	10 Hours	CO4
Total Hours		40 Hours	

Self Learning Module:

20 HOURS

Objective:

To encourage self-directed learning and enhance knowledge of Java programming concepts through online platforms, coding practice, video tutorials, and additional reading resources.

Recommended Online Resources:

Sr. No.	Unit Name/Topic Name	Resource Link
1	Java Basics, OOP Concepts, JVM, Classes and Objects	1. College Recorded Videos 2. NPTEL / MOOC / SWAYAM 3. YouTube Tutorials 4. Various Reading Materials and E-books 5. Online Coding Platforms and Documentation
2	Arrays, Inheritance, Interface, Exception, Handling, Generic Programming	1. Oracle Java Documentation 2. NPTEL Courses 3. SWAYAM Portal 4. W3Schools Java Tutorial 5. JavaTpoint Java Tutorial

CO-PO-PSO Matrix (For Theory):

Co. No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2
CO-1	3	2	-	-	-	2	1	2	2
CO-2	3	3	2	2	2	-	-	3	2
CO-3	3	3	2	-	2	2	2	3	3
CO-4	3	3	2	2	3		2	3	2

CO-PO-PSO Matrix (For Practical): N.A**CO-PO-PSO MAPPING JUSTIFICATION FOR CORE JAVA (For Theory):**

CO No.	CO Statement	PO No.	Justification
CO-1	Explain core Java concepts including object-oriented principles, program structure, data types, control flow, classes, objects, constructors, arrays, and command-line arguments.	PO1	Applies fundamental computing and programming knowledge to understand Java syntax, data types, control structures, arrays, classes, objects, and object-oriented concepts.
		PO2	Analyzes basic programming problems and identifies suitable Java constructs, data types, control structures, and object-oriented techniques to solve them.
		PO6	Develops awareness of responsible and reliable programming practices while designing basic Java applications for practical use.

		PO7	Encourages awareness of efficient and sustainable software practices through reusable Java programs and effective utilization of computing resources.
		PSO1	Develops foundational Java programming and object-oriented skills required for software application development.
		PSO2	Builds fundamental problem-solving and programming competencies required to develop technology-based software solutions.
CO-2	Apply inheritance, polymorphism, interfaces, packages, strings, wrapper classes, and related concepts to develop reusable object-oriented Java programs.	PO1	Applies knowledge of object-oriented programming concepts such as inheritance, polymorphism, abstraction, interfaces, packages, and standard Java classes.
		PO2	Analyzes programming requirements and selects appropriate inheritance, interfaces, packages, and object-oriented techniques for solving problems.
		PO3	Designs reusable and extensible Java solutions using class hierarchies, inheritance, polymorphism, abstract classes, and interfaces.
		PO4	Investigates the behavior of inheritance, method overloading, method overriding, access control, and object interactions through program execution and experimentation.
		PO5	Utilizes Java development environments, standard APIs, packages, and debugging tools for developing and testing object-oriented applications.
		PSO1	Strengthens object-oriented software development skills through practical implementation of inheritance, interfaces, packages, and reusable Java components.
		PSO2	Applies object-oriented design techniques to develop modular, maintainable, and extensible software solutions.
CO-3	Apply exception handling and multithreading concepts to develop robust and concurrent Java applications.	PO1	Applies programming knowledge to understand exception handling, exception hierarchy, threads, thread creation, and concurrent execution in Java.
		PO2	Analyzes runtime errors and concurrent programming requirements to select appropriate exception-handling and multithreading techniques.
		PO3	Designs robust and concurrent Java applications using structured exception handling, user-defined exceptions, and multithreading concepts.

		PO5	Uses Java APIs, thread classes, exception-handling mechanisms, IDEs, and debugging tools to implement and test reliable applications.
		PO6	Promotes responsible software development by handling errors effectively and improving the reliability and stability of Java applications.
		PO7	Encourages efficient use of computing resources through concurrent execution and effective thread management.
		PSO1	Enhances Java application development capabilities by implementing robust exception-handling and concurrent programming techniques.
		PSO2	Applies advanced Java programming techniques to develop reliable, responsive, and efficient software solutions.
CO-4	Apply generics, collections, utility classes, regular expressions, and related Java features to develop flexible, reusable, and efficient programs.	PO1	Applies knowledge of generics, collections, utility classes, and standard Java APIs to develop structured and efficient programs.
		PO2	Analyzes application requirements and selects appropriate generic types, collections, and utility classes for effective data processing and management.
		PO3	Designs flexible and reusable software solutions using generic classes, generic methods, List, Set, Map, and other Collection Framework components.
		PO4	Investigates the behavior and performance of different collection types, generic structures, regular expressions, and utility classes through experimentation.
		PO5	Effectively utilizes Java Collection Framework, standard APIs, IDE features, and development tools for implementing modern Java applications.
		PO7	Promotes efficient utilization of computing resources by selecting suitable data structures, reusable generic components, and optimized collection mechanisms.
		PSO1	Strengthens advanced Java programming skills through practical use of generics, collections, regular expressions, and standard utility APIs.
		PSO2	Applies modern Java programming techniques to develop flexible, reusable, efficient, and scalable software solutions.

CO-PO-PSO MAPPING JUSTIFICATION (For Practical): NA

MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS

SDG Mapping for CORE JAVA(CAM5001C)		
SDG No and Title	Relevance to Course	Justification
SDG 4: Quality Education	High	Core Java equips students with fundamental programming and problem-solving skills, enhancing digital literacy and employability in the IT sector.
SDG 8: Decent Work and Economic Growth	High	Java programming skills prepare students for careers in software development, contributing to workforce readiness and economic growth.
SDG 9: Industry, Innovation and Infrastructure	High	The course develops competencies in software application development, fostering innovation and supporting digital infrastructure development.
SDG 10: Reduced Inequalities	Moderate	Open-source Java tools and learning resources provide equal opportunities for students to acquire programming skills regardless of economic background.
SDG 11: Sustainable Cities and Communities	Moderate	Java can be used to develop smart applications for urban services, resource management, and community-focused digital solutions.
SDG 17: Partnerships for the Goals	Moderate	Collaborative programming projects and the use of open-source technologies encourage knowledge sharing and teamwork among learners and developers.

Explanation:

The Core Java course supports sustainable development by providing students with fundamental programming knowledge, computational thinking, and software development skills required in the

digital era. The course enables learners to design and develop efficient software applications using object-oriented programming principles, thereby enhancing their technical competencies and employability. Through practical problem-solving, application development, and the use of modern programming tools, students are prepared to contribute to innovation, digital transformation, and technology-driven solutions for societal and industrial challenges. The course primarily aligns with SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth), and SDG 9 (Industry, Innovation and Infrastructure) while also supporting inclusive learning, collaborative development, and sustainable technological advancement.